

Multi-targeted prevention of cancer by sulforaphane

Isothiocyanates are found in cruciferous vegetables such as broccoli, Brussels sprouts, cauliflower, and cabbage. Epidemiologic studies suggest that cruciferous vegetable intake may lower overall cancer risk, including colon and prostate cancer. Sulforaphane (SFN) is an isothiocyanate found in cruciferous vegetables and is especially high in broccoli and broccoli sprouts. SFN has proved to be an effective chemoprotective agent in cell culture, carcinogen-induced and genetic animal cancer models, as well as in xenograft models of cancer. Early research focused on the “blocking activity” of SFN via Phase 2 enzyme induction, as well as inhibition of enzymes involved in carcinogen activation, but there has been growing interest in other mechanisms of chemoprotection by SFN. Recent studies suggest that SFN offers protection against tumor development during the “post-initiation” phase and mechanisms for suppression effects of SFN, including cell cycle arrest and apoptosis induction are of particular interest. In humans, a key factor in determining the efficacy of SFN as a chemoprevention agent is gaining an understanding of the metabolism, distribution and bioavailability of SFN and the factors that alter these parameters. This review discusses the established anti-cancer properties of SFN, with an emphasis on the possible chemoprevention mechanisms. The current status of SFN in human clinical trials also is included, with consideration of the chemistry, metabolism, absorption and factors influencing SFN bioavailability.